

HAI 2025/2026

Beatriz Guerreiro

Francisco Guimarães

Sofia Reia

Option A

GUARD

Guidance for **U**ser **A**wareness, **R**easoning and **D**ependence)

Problem Description

User over-reliance on LLM based systems

- Most people trust AI systems too much, lacking critical thinking and rarely checking sources
- AI information may be outdated or generated and the user may not notice it
- Users rarely verify AI outputs before acting

Current systems don't detect or interrupt this

Unstable Intelligence: GenAI Struggles with Accuracy and Consistency

Mesut Cicek

Washington State University, USA

Sevincgul Ulu

Southern Illinois University, USA

Can Uslay

Rutgers University, USA

Kate Karniouchina

Northeastern University, USA

Abstract

This study examines the accuracy and consistency of Generative AI (GenAI) by testing ChatGPT's ability to estimate the accuracy of 719 business research hypotheses. For critical tasks, we find GenAI performance to be inadequate in terms of accuracy and consistency. Accuracy improved only marginally from 76.5% (GPT-3.5, 2024) to 80% (GPT-5 mini, 2025), yielding an effective chance-adjusted accuracy of only 60%. Moreover, accuracy drops significantly for insignificant hypotheses, reaching only 16.4% in 2025. Crucially, consistency across ten identical prompts was poor, with over a quarter of the cases having at least one incorrect estimation. We conclude that GenAI's linguistic fluency is not yet backed by commensurate conceptual intelligence and frequently produces unreliable output, necessitating vigilant human oversight.

Mesut Cicek, Sevincgul Ulu, Can Uslay, Kate Karniouchina (Fall 2025). **Unstable Intelligence: GenAI Struggles with Accuracy and Consistency.**

Target Users

University Students

Students that want help studying but do not want to blindly trust in LLMs and AI systems

People who casually use and blindly trust LLMs/AI systems

So they can be reminded that not all information is completely correct, that LLMs make mistakes and have the ability to generate fake sources

Workers in IT and management

Analysts, consultants, marketers, developers that may already use AI in their day-to-day work and could use an extra layer of assurance and credibility

System Concept

An AI-powered interface that **runs on top of a chatbot/LLM** and monitors user interaction to detect over-reliance and intervenes with explanations, sources and verification prompts:

1. Tracks interaction patterns (questions, follow-ups, time-to-action)
2. Estimates a **trust score** (from healthy Collaboration to **blind trust**)
3. Triggers interventions when risky patterns are detected

System Concept

Behavioral signals of over-reliance

We infer trust from the user's **observable behaviors**:

- **Low verification**
 - No follow-up questions
 - No request for sources and minimal interaction with the recommended sources by the system
- **Fast action**
 - Immediate copy / execution
- **Prompt Monotony**
 - Lack of alternative queries or variety in their requests

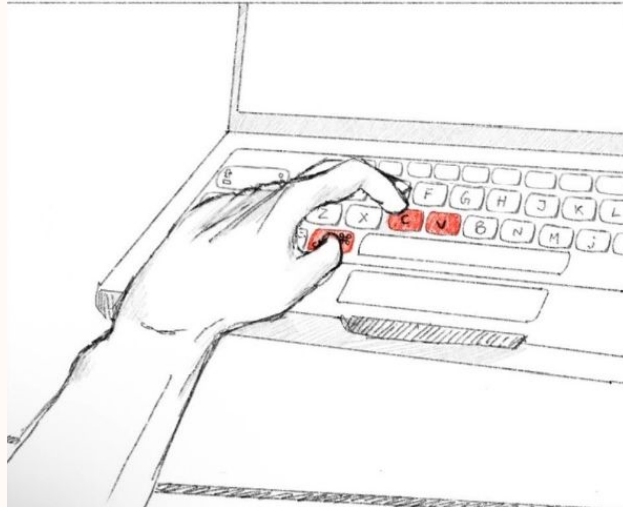
Combine into a **Blind Trust Score**

Blind Trust in AI

Justin is rushing to finish his project.
He opens a standard AI chat,
looking for a shortcut.



He copies and pastes the AI's
response without a second thought.
No fact-checking, no questions asked...



The information was outdated or
hallucinated.
Justin stares at the screen in disbelief.
He failed the course.



Healthy Trust in AI

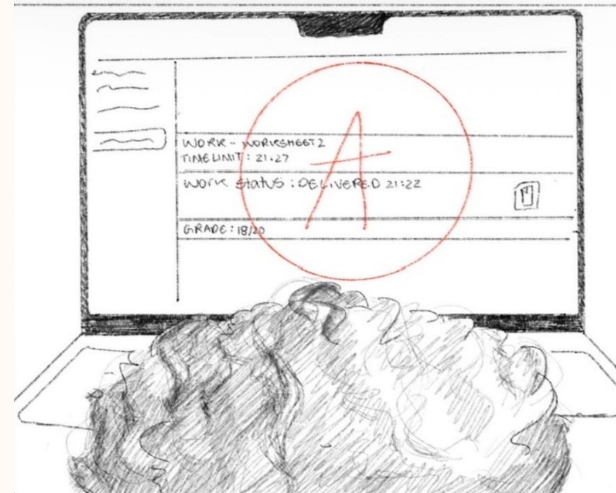
Next time, Justin tries a different tool: **GUARD**. He wants to be a collaborator, not just a spectator.



GUARD provides a trust Rate and specific sources. Justin notices a low-confidence warning and decides to dive deeper into the recommended articles.



The final work is accurate and insightful. Justin knows he did a good work with the help of **GUARD**. He got an A in the project.



Data Ecosystem

- **Data Required:** User inputs, interaction behavior (clicks, follow-ups, time), and user feedback

- **Data Source:** Direct interaction between the user and the system

- **Processing:** The AI model generates responses and analyzes behavior to estimate a “trust score”

- **Output:** AI response, confidence level, and alerts encouraging verification when needed

- **Improvement Over Time:** The system adapts based on user feedback and interaction history

- **User Influence:** The users can question answers, request sources, or provide corrections

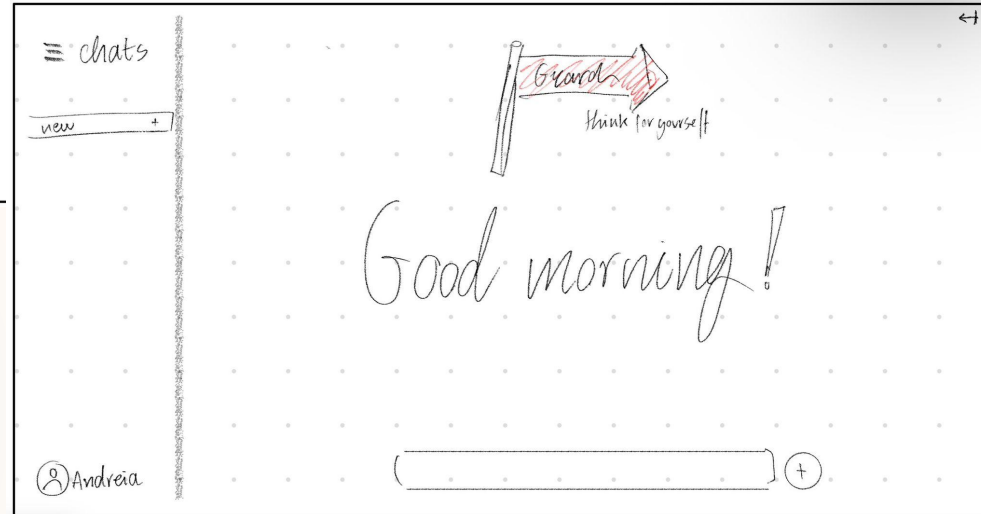
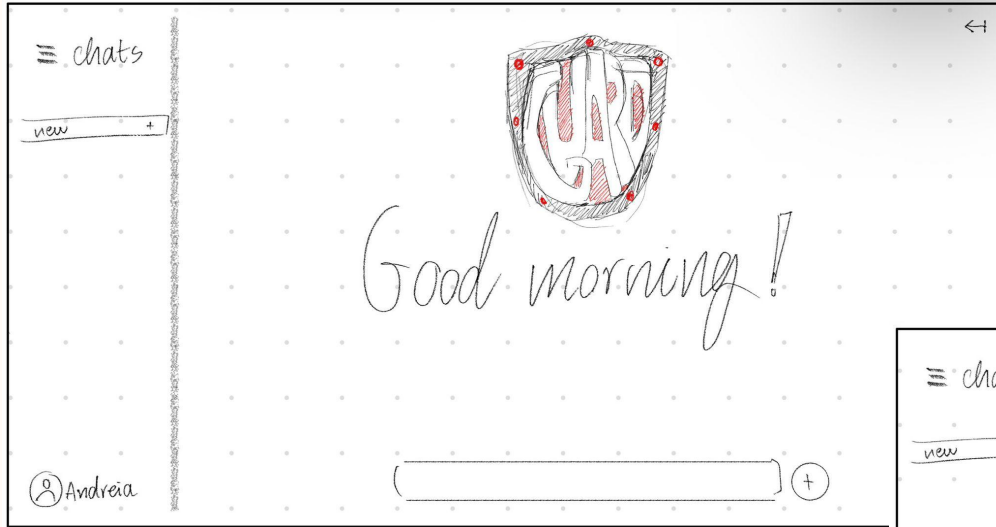
- **Risks:** Bias in interpreting behavior, and difficulty in accurately assessing user trust and model confidence

Initial Design of the AI Behavior

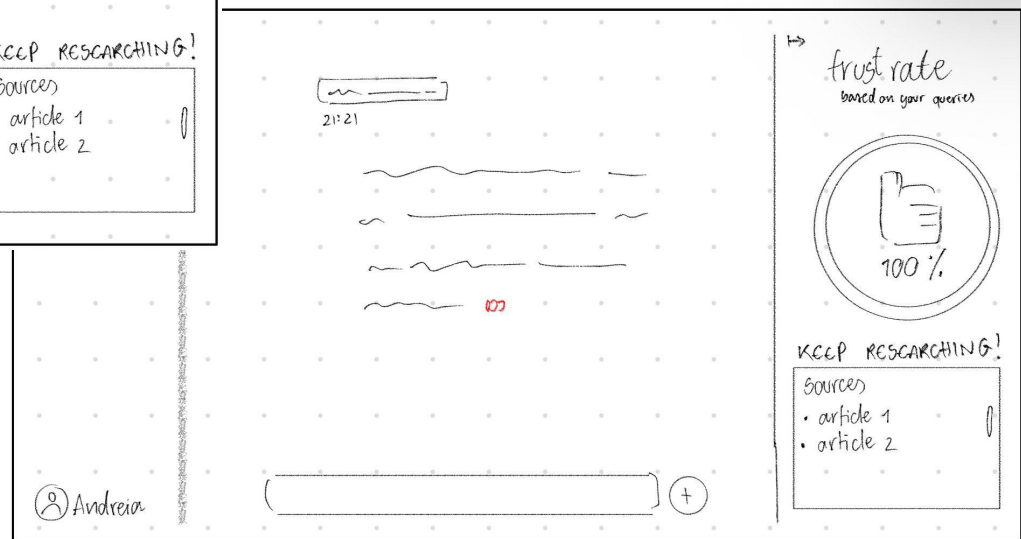
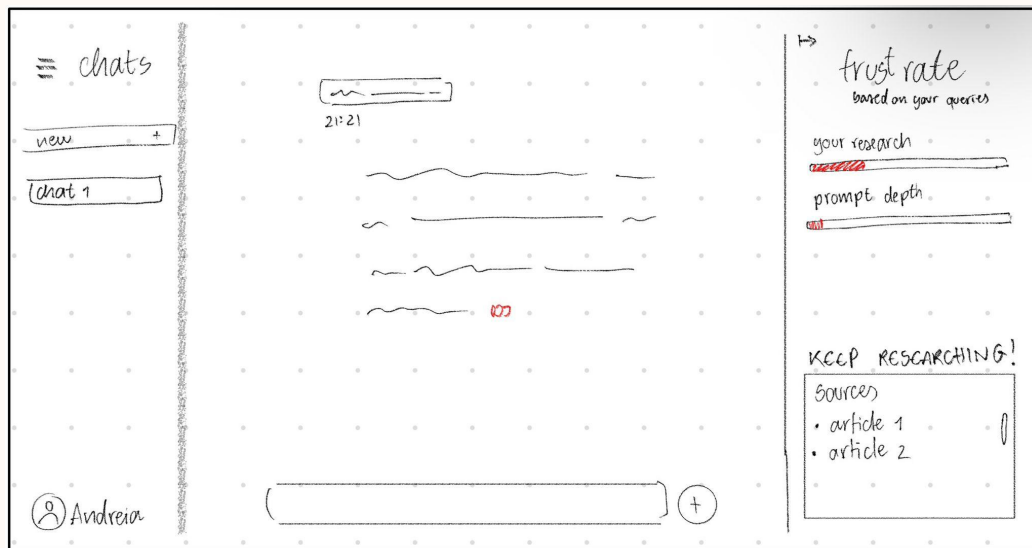
By relying on **behavioral data** and **interaction history**, the AI takes the initiative to interrupt risky over-reliance with overlay explanations, verification prompts, and a visible score based on the user's queries.

To account for its own imperfections, the system **handles uncertainty by flagging outdated or hallucinated information**, encouraging the user to act as a *collaborator* who validates sources rather than a *passive spectator*.

Initial Design of GUARD Interface



Initial Design of GUARD Interface





**Any
Questions?**



CREDITS: This presentation template was created by **Slidesgo**, and includes icons by **Flaticon** and infographics & images by **Freepik**

Thanks!

